order22 RESULTS
PREREG sha256 f7ca80834e5c3ddf5a1b454f69164ac52451ac45f600381d06a7039b31cf0982 (hashed before numerics; Maps RG9, RG9b hashed there).
test	value	verdict
C1 general corner formula on (e1,e2) reproduces A2_corner	max abs 0.0	PASS
C2 Γ·G = G (γ-invariance of the Gauss means)	1.8e−16	PASS
Map RG9: Γ·A2_corner = Corner_{γ⁻¹K}	e9 = 5.6e−15; non-vacuity ‖Γ·A2c − A2c‖/‖A2c‖ = 1.60	**PASS (non-vacuous)**
Map RG9b: Corner(e1,(1,1)) + Corner((1,1),e2) = Corner(e1,e2)	max abs 0.250	**FAIL**

Reading
- RG9: the unit γ does not preserve the corner term (defect 160% of its norm), and it moves it exactly to the corner invariant of the transported cone γ⁻¹K = cone((13,−2),(7,−1)). The γ-defect of A2_corner is therefore entirely geometric: Γ·A2c − A2c = Corner_{γ⁻¹K} − Corner_K.
- RG9b FAIL is recorded by rule. Post-hoc explanation (not a pass): the preregistered additivity ignored the vertex weight. Each subcone carries weight ¼ at ν = 0, so the split puts ½ at the origin against ¼ for the quadrant; the extra ¼·f(0) on companions with α ≡ 0 gives exactly the observed 0.25. The corrected statement (subtract ¼·[0 ∈ α+Z²]·f(0)) must be re-preregistered before it counts.

Category A
- Transport identity S^E_{γα,γβ} = S^{E∘γ}_{α,β} ⇒ Γ·Corner_K = Corner_{γ⁻¹K} (substitution plus reduction rules); the orthant arc is exactly half the unit arc (cosh L = √(7/2)).
- The general unimodular-cone corner formula (coefficients B(u1,u2)/(4Q(u2)), B(u1,u2)/(4Q(u1))) by the iterated Euler–Maclaurin argument of paper v6.

Category B
Numerical confirmations above.

order22 NOTES
- Lock respected: MAPS_CANDIDATES untouched; no Π / ±1 grading / path predicate; no Resonant object. RG9 is residual-internal geometry; its pass is not a Residual↔Resonant bridge and supplies no Resonant candidate.
- RG9 is close to an identity once the transport lemma is granted; its value is the non-vacuity (γ moves A2_corner by 160%) and the confirmation of the general cone formula with phases and reductions.
- RG9b FAIL: vertex-weight omission in my preregistered statement (see RESULTS). Not re-run.
- Near-law (order20/21): not explained here. A geometric candidate for order23 (to preregister): average A2_corner over the γ-orbit of cones covering the unit arc (two orthant arcs = one unit arc), i.e. Corner_K + Corner_{σK} for the reflection σ fixing ĉ′, and test whether that γ-averaged corner obeys the w=1 law.
- v8 not built: the new Category A content (transport, general cone corner) is short and better folded with a corrected RG9b.
- Ξ not identified. E1–E3 OPEN.
- nice, OMP_NUM_THREADS=1, <3 GB, ~25 s; no push/sends; new files only.